

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONTACT                    | Email: <a href="mailto:sunayj@sas.upenn.edu">sunayj@sas.upenn.edu</a><br>Website: <a href="https://sunayjoshi.github.io">sunayjoshi.github.io</a><br>GitHub: <a href="https://github.com/sunayjoshi">github.com/sunayjoshi</a><br>Google Scholar: <a href="https://scholar.google.com/citations?user=8fTMwcUAAAAJ">scholar.google.com/citations?user=8fTMwcUAAAAJ</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| RESEARCH INTERESTS         | Statistical machine learning; graphical and network models; high-dimensional asymptotic statistics; statistical uncertainty quantification; applied selective inference; foundations of LLM post-training; applied stochastic optimal control; applications to financial econometrics and macroeconometrics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| EDUCATION                  | <b>University of Pennsylvania</b> , Ph.D. in Applied Mathematics 2024–2028 (expected) <ul style="list-style-type: none"> <li>Advisors: Edgar Dobriban and Hamed Hassani</li> <li>Committee: Edgar Dobriban, Hamed Hassani, Zhimei Ren, Aaron Roth</li> <li>Title: “Conformal and Selective Inference for Reliable AI”</li> </ul> <b>Princeton University</b> , A.B. in Mathematics with Highest Honors ( <i>summa cum laude</i> ) 2020–2024 <ul style="list-style-type: none"> <li>Minors: Applied and Computational Mathematics; Applications of Computing</li> <li>Senior Thesis Advisor: Allan Sly</li> <li>Title: “Alignment of Correlated Random Graphs”</li> <li>GPA: 3.98/4.00</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| PUBLICATIONS AND PREPRINTS | <p>Three working papers on (1) structure-aware conformal factuality via graphical models; (2) statistical proof verification via multiple hypothesis testing; (3) finite-sample theory of synthetic data for estimating linear models. (All to be submitted to ICLR 2027 in September 2026.)</p> <p><b>Sunay Joshi</b>, Tao Wang, Hamed Hassani, and Edgar Dobriban. “Risk-Controlled Post-Processing of Decision Policies.” <i>NeurIPS</i>, 2026. <a href="https://arxiv.org/abs/2605.06479">arXiv:2605.06479</a>.</p> <p><b>Sunay Joshi</b>, Yan Sun, Hamed Hassani, and Edgar Dobriban. “MultiRisk: Multiple Risk Control via Iterative Score Thresholding.” <i>Revise and resubmit at JASA</i>. <a href="https://arxiv.org/abs/2512.24587">arXiv:2512.24587</a>, 2025.</p> <p>Georgy Noarov, Soham Mallick, Tao Wang, <b>Sunay Joshi</b>, Yan Sun, Yangxinyu Xie, Mengxin Yu, and Edgar Dobriban. “Foundations of Top-<math>k</math> Decoding for Language Models.” <i>NeurIPS</i>, 2025. <a href="https://arxiv.org/abs/2505.19371">arXiv:2505.19371</a>.</p> <p><b>Sunay Joshi</b>, Shayan Kiyani, George J. Pappas, Edgar Dobriban, and Hamed Hassani. “Conformal Inference under High-Dimensional Covariate Shifts via Likelihood-Ratio Regularization.” <i>NeurIPS</i>, 2025. <a href="https://arxiv.org/abs/2502.13030">arXiv:2502.13030</a>.</p> <p>Gary Hu, Julian Gold, Uthsav Chitra, <b>Sunay Joshi</b>, and Benjamin J. Raphael. “Anomaly Detection in Spatial Transcriptomics via Spatially Localized Density Comparison.” <i>Bioinformatics</i>, 41(Supplement 1):i493–i501, 2025. doi:10.1093/bioinformatics/btaf242.</p> <p>Marissa Gee, Nicolas Gonzalez-Granda, <b>Sunay Joshi</b>, Nagaprasad Rudrapatna, Anne Somalwar, Stephen P. Ellner, and Alexander Vladimirovsky. “Navigating the Landscape of Fear.” <i>bioRxiv</i>, 2024. doi:10.1101/2024.08.15.608007.</p> |
| TALKS AND PRESENTATIONS    | <ul style="list-style-type: none"> <li><b>Poster Presentation (upcoming)</b>, NeurIPS Conference 2026<br/>“Risk-Controlled Post-Processing of Decision Policies.”</li> <li><b>Poster Presentation</b>, STAI-X Conference, Harvard University August 2026<br/>“Risk-Controlled Post-Processing of Decision Policies.”</li> <li><b>Poster Presentation</b>, NeurIPS Conference, San Diego December 2025<br/>“Conformal Inference under High-Dimensional Covariate Shifts via Likelihood-Ratio Regularization.”</li> <li><b>Virtual Talk</b>, EnCORE Collaboration Workshop, UC San Diego September 2025<br/>“Conformal Inference under High-Dimensional Covariate Shifts via Likelihood-Ratio Regularization.”</li> <li><b>Group Talk</b>, Dobriban Stat-ML Reading Group, University of Pennsylvania January 2026<br/>“Introduction to LLM reasoning.”</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## TEACHING

- **Undergraduate Course Assistant**, Princeton University    September 2023–December 2023
  - Served as one of two undergraduate assistants for MAT 340: Applied Algebra.
  - Led weekly problem sessions on introductory group theory and representation theory.

## REVIEWING

- Reviewer: *Journal of the American Statistical Association* (JASA), 2025; STAI-X 2026; NeurIPS 2026; OPT Workshop at NeurIPS 2026; ICML 2026 Workshop on AI4Research.

## SERVICE AND OUTREACH

- **Volunteer**, ICML 2026 Workshop on AI4Research    2026
- **Mentor**, Directed Reading Program (DRP)    Spring 2026, Fall 2026
  - Topics: high-dimensional statistics, distribution-free statistics.
- **Organizer**, Dobriban Stat-ML Reading Group    September 2024–April 2025
  - Organized logistics for Prof. Edgar Dobriban’s statistics and machine learning reading group.
  - Selected speakers for weekly meetings, managed shared materials, and made announcements.
- **Volunteer**, University of Pennsylvania Math Circles    September 2024
  - Provided after-school mathematics enrichment activities for students at West Philadelphia High School.
- **Advising Chair**, Princeton Mathematics Club    April 2023–May 2024
  - Organized course advising sessions for undergraduates in the mathematics department.
  - Organized the Mentoring Möbius program connecting undergraduates with mathematics graduate students.
- **Problem Czar**, Princeton Math Contest    2022
  - Served as lead problem writer for Princeton’s annual high school mathematics contest (PUMaC).
  - Held regular problem-writing meetings, selected final contest problems, and wrote 48 problems over four years.

## AWARDS AND HONORS

- Sigma Xi, 2024: elected to national science honor society.
- Phi Beta Kappa, 2024: elected to the oldest academic honor society in the United States.
- NSF Graduate Research Fellowship Program Honorable Mention, 2024.
- Andrew H. Brown Prize, 2023: awarded by the Princeton Mathematics Department to outstanding juniors in mathematics.
- Class of 1861 Prize, 2022: awarded by Princeton to the sophomore with the best record on the Putnam Examination.
- Shapiro Prize for Academic Excellence, 2021: recognizes outstanding academic achievement by Princeton undergraduates in their first or second years of study.
- Rank 150, William Lowell Putnam Mathematical Competition, 2021.
- Silver Medalist, USA Physics Olympiad, 2019.
- Qualifier, USA Junior Math Olympiad, 2018.

## TECHNICAL SKILLS

- *Programming*: Python, Java, C++, Linux, Bash, GitHub, agentic programming workflows.
- *Typesetting*: LaTeX.
- *Languages*: English (native), Spanish (intermediate), Hindi (beginner).